

MOVIE SENTIMENT ANALYSIS: NLP PIPELINE

Deep Learning Text Classification on 50,000 IMDB Reviews

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Domain: Deep Learning NLP

Stack: PyTorch, BiLSTM, NLTK

PERFORMANCE METRICS

89.0%

Classification Accuracy

+4.5%

vs Classical Baseline

50,000

IMDB Movie Reviews

BiLSTM

Neural Network

PIPELINE ARCHITECTURE

This text classification system processes 50,000 IMDB movie reviews through tokenization, lemmatization, and stop word removal before passing feature representations to a PyTorch BiLSTM neural network.

- Dataset Source: Kaggle IMDB Dataset of 50K Movie Reviews
- Local Data Cache: Included in data/imdb_reviews_sample.csv
- Model Accuracy: 89.0% test accuracy, beating Logistic Regression and Random Forest models